

Security Two

Project Assignment: Secure Messaging Application

The objective of this project is to apply the cryptographic concepts studied in the course, particularly **hash functions** and **encryption techniques**, by implementing a simple secure messaging web application.

Students will design and implement a system that allows users to create accounts and exchange encrypted messages. The project demonstrates how cryptographic techniques are used to protect **user credentials** and **message confidentiality**.

The implementation language for this project must be **JavaScript (JS)**.

Technology Requirements

Students must implement the project using **JavaScript**.

The following libraries should be used:

- **bcrypt** for password hashing
- **node-forge** for RSA key generation and encryption

Libraries:

- bcrypt
- node-forge

Students may use **Node.js** for the backend and any simple frontend framework (or plain HTML/JS).

Project Description

Students are required to develop a **web-based messaging application** in which users can register, log in, and send messages to other users in the system.

The application must ensure that **user passwords are securely stored** using hashing, and that **messages are encrypted before being transmitted or stored**.

Each student is free to choose the encryption approach used in their implementation.

Two possible cryptographic approaches are suggested:

1. **Symmetric Encryption using Caesar Cipher**
2. **Asymmetric Encryption using RSA**

Students must choose **one of the two approaches** when implementing the encryption system.

Functional Requirements

1. User Registration

The system must allow new users to create accounts using:

- Username
- Password

Passwords must **not be stored in plain text**. Instead, they must be stored as a **hashed value**.

Example:

```
stored_password = hash(password)
```

Students must use the **bcrypt** library to hash passwords before storing them in the database.

2. User Authentication (Login)

The system must allow registered users to log in.

During login:

1. The entered password is hashed.
2. The resulting hash is compared with the stored hash.

Access is granted only if the hashes match.

3. Messaging System

Users must be able to send messages to other registered users.

Messages must be **encrypted before being stored or transmitted**.

Students must choose **one of the following encryption approaches**.

Option A: Symmetric Encryption (Caesar Cipher)

Students may implement a **Caesar Cipher encryption system**.

The message is encrypted using a **shift value**.

Example:

HELLO → KHOOR

The same shift value is used for both:

- Encryption
 - Decryption
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Option B: Asymmetric Encryption (RSA)

Students may implement **RSA encryption** using the **node-forge** library.

In this approach:

- Each user has a **public key** and a **private key**.
- Messages are encrypted using the **receiver's public key**.
- Messages are decrypted using the **receiver's private key**.

Only the intended receiver should be able to read the message.

Suggested Database Structure

Users Table

Field	Description
id	User identifier
username	User name
password_hash	Hashed password
public_key	Public key (if RSA is used)

Messages Table

Field	Description
id	Message identifier
sender_id	Sender user ID
receiver_id	Receiver user ID
ciphertext	Encrypted message
encryption_type	Encryption method used
timestamp	Message time

Expected Features

The application should include:

- User registration
 - Secure password hashing using bcrypt
 - Login authentication
 - Sending encrypted messages
 - Storing encrypted messages in the database
 - Decrypting messages when received
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Submission Requirements

Students must submit:

- Complete **source code**
- A **README file** explaining:
 - How to run the project
 - Which encryption method was used
 - How password hashing was implemented

Evaluation Criteria

Component	Marks
User Registration & Password Hashing	3
Login Authentication	2
Encryption Implementation	10
Messaging Functionality	3
Code Organization & Documentation	2
Total: 20 Marks	